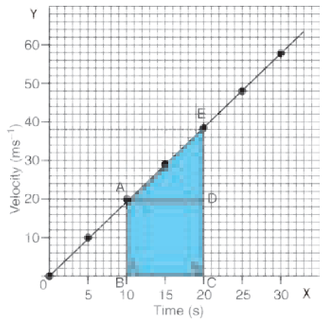


Figure 2.6 Velocity – Time graph for uniform accelaration

One can also determine the distance moved by the car from its velocity – time graph. The area under the velocity – time graph gives the distance (magnitude of displacement) moved by the car in a given interval of time.

Since the magnitude of the velocity of the car is changing due to acceleration, the distance, S travelled by the car will be given by the area ABCDE under the velocity – time graph. That is,

S = Area ABCDE

= Area of the rectangle ABCD + Area of the triangle ADE

$S = (AB \times BC) + \frac{1}{2} (AD \times DE)$

Area of the quadrangle ABCDE can also be calculated by calculating the area of trapezium ABCDE. It means,

S = Area of trapezium ABCDE

= ½ × Sum of length of parallel sides × Distance between parallel sides

$S = \frac{1}{2} \times (AB + CE) \times BC$

In the case of non-uniformly accelerated motion, distance – time graph and velocity – time graphs can have any shape as shown in Figure 2.7.

DO YOU KNOW?

The speedometer of an automobile measures the instantaneous speed of the automobile. In a uniform motion in one dimension, the average velocity is equal to instantaneous velocity. Instantaneous velocity is also called velocity or instantaneous speed or simply speed.

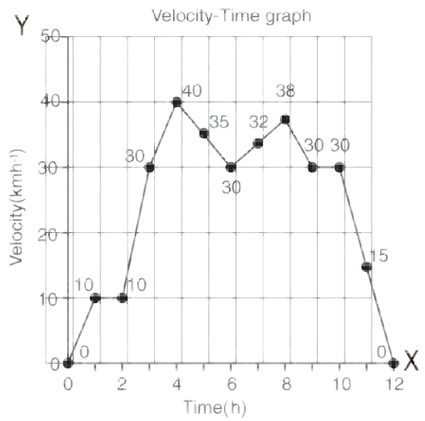


Figure 2.7 Velocity – Time graph for Non-uniform accelaration

2.6 Equations of Motion

Newton studied the motion of an object and gave a set of three equations. These equations relate displacement, velocity, acceleration and time of an object under motion. An object in motion with initial velocity, u attains a final velocity, v in time, t due to acceleration, a and reaches a distance, s. Three equations can be written for this motion.

$$v = u + at$$

$$s = ut + \frac{1}{2} a t^2$$

$$v^2 = u^2 + 2as$$

Let us try to derive these equations by graphical method.

Figure 2.8 shows the change in velocity with time for an uniformly accelerated object. The object starts from the point D in the graph with velocity, u. Its velocity keeps increasing and after time, t it reaches the point B on the graph.

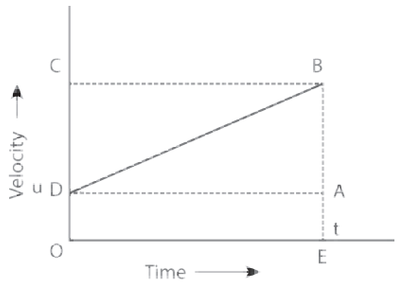


Figure 2.8 Equations of Motion

The initial velocity of the object = u = OD = EA

The final velocity of the object = v = OC = EB

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